IAE PARIS — SORBONNE BUSINESS SCHOOL · IDGP ACADEMY

Digital Tools & Benchmarking

Reading notes for the afternoon session. Module 3 — Methods, Tools and Operational Efficiency, Session 4.

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Prof. Dr. Thiago Lima

International Certificate in Project Management and Organizational Performance

No preparation is required. The afternoon tests the morning's argument against the two things organisations buy when they want capability and cannot build it: technology, and other people's practices.

Contents

What a digital investment actually buys; what an AI system was never told and what nobody measured; why best practice resists transfer; and how a transformation is run when the path cannot be known in advance.

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> **The instrument.** The group activity in Part Seven is run on a separate booklet, the *Digital Tool Appraisal*, which is fillable online and printed here as a companion. Bring one tool you are evaluating, or one you have already bought and are not sure about.

This morning ended on a claim: value is not the sum of what an organisation owns and how it works, but that sum scaled by its capacity to improve. The afternoon tests the claim against the two things organisations buy when they want that capacity and cannot build it — technology, and other people's practices.

The order is deliberate. Digital platforms come first because they are the most expensive thing on the list and the easiest to mistake for capability. Artificial intelligence comes next because it is the extreme case of the same error. Benchmarking comes last because it is the only one that looks like it should work.

> **WHAT TO BRING FROM THE MORNING**
>
> **Your Value Case canvas** — the project your group appraised. This afternoon examines the tooling that project depends on.
>
> **Your NUMMI hypothesis**, and the resolution we reached. The same mechanism explains why buying enterprise software does not build competitive advantage, and the puzzle below is its digital twin.

ONE

The digital puzzle

Two firms, one platform

The setting. Two organisations in the same industry, of comparable size, buy the same enterprise platform from the same vendor in the same year. Same modules, same implementation partner, comparable budgets. One of them also deploys the same demand-forecasting model, trained on a similar history.

Three years later, one has materially shorter lead times and a forecast its planners trust. The other has the platform, the licences, the dashboards — and a planning process that works around all three.

Nothing in the purchase differed. **Where did the difference come from, and why was it invisible in both business cases?**

If this sounds like Fremont with better software, that is the point. The morning established that assets and processes are additive and purchasable while capability is a coefficient. This afternoon the assets are more expensive, the vendors more persuasive, and the arithmetic unchanged.

TWO

What a digital investment actually buys

Almost every platform is an **A** decision. Some are also a **P** decision. Very few are a **C** decision — and the error that destroys the business case is not buying A, it is booking the benefit as though C had moved.

$$V = C \times (A + P)$$

a platform enters the bracket · the coefficient is bought separately, or not at all

The landscape, in six categories

The market is large, it changes faster than any syllabus, and nothing below is a recommendation. The categories are what matter, because **the category tells you what a tool does and the VCAP term tells you what it is worth**. The named products are examples only, offered so the categories are recognisable.

Category	Examples	What it typically changes	Usual term
Collaboration and work management	Asana, Monday, Trello, Planner, Jira	Visibility of tasks and ownership	A, and P if the flow of work is genuinely redesigned
Scheduling and portfolio	MS Project, Primavera P6, Planview, Clarity	Baselines, dependencies, resource and portfolio views	A, occasionally P
Analytics and dashboards	Power BI, Tableau, Looker	What leadership can see, and how often	A — and see the caution below
Integration and the data layer	APIs, integration platforms, a warehouse or lakehouse	Where the authoritative version of each fact lives	Frequently the only genuine P in a programme
Automation and workflow	Workflow engines, robotic process automation	Steps performed without a person	P, and A
AI and assistive	Assistants, forecasting models, document generation, agents	Predictions, drafts, and increasingly actions	A — Parts Three and Four are about why

Two observations that survive every change in the market. The least visible category — integration and the data layer — is the one that most often carries the real process change, and it is the one hardest to put in a business case because it demonstrates nothing on a screen. And the most visible category, dashboards, changes what is seen without changing what is done, unless somebody also changes the decision the dashboard feeds.

> **A CAUTION ON DASHBOARDS, CARRIED OVER FROM THIS MORNING**
>
> A dashboard reports variation; it does not distinguish the two kinds. A governance routine that demands an explanation for every movement against target — favourable or adverse — is treating common-cause variation as though it were special cause, which this morning established makes the process worse rather than better. The instrument is not the problem. The routine attached to it is, and the routine is a **P** decision that almost never appears in the tooling business case.

The triage, applied to a tool

Term	What a tool contributes	The test
A — Assets	Licences, infrastructure, models, dashboards. The organisation owns something it did not own before.	Could a competitor with the same budget hold this position in six months? Usually yes.
P — Processes	The flow of work changes: approvals route differently, handoffs disappear, a queue is capped.	Would the new flow survive if the tool were replaced by a different vendor's?
C — Capability	The organisation ends up better at improving its processes than it was before.	What can it now do that it could not do before — *that does not depend on the new system?*

Training people to operate the new system is not capability. It is part of delivering the asset, and it ends when the people who were trained leave. The C test is deliberately hard to pass, because most honest answers are A, or A and P, and recording that truthfully is what makes the appraisal useful.

The productivity paradox, and why it is not about technology

Research on information technology investment has repeatedly found the same shape: the returns to technology appear where complementary organisational practices exist, and are close to nil where they do not. The pattern survived decades of changing technology because it was never about the technology.

It is the same complementarity argument from this morning, in a different literature. A practice adopted without its complements imposes its costs and withholds its returns — and a platform is a practice that arrives with an invoice attached.

IN PROJECT TERMS

Two consequences follow immediately, and most portfolio platform business cases contain neither.

Adoption is the return. A tool used as the working record by 40% of the people it was bought for does not deliver 40% of a single-source-of-truth benefit. It delivers close to none, because the remaining 60% means every figure still has to be reconciled. Budget change management with the same seriousness as licences, or expect the arithmetic to punish you.

One source of truth per data type. A schedule date, a risk, a cost figure: each needs exactly one authoritative home, with everything else consuming it. The moment the same fact lives in two systems the organisation starts paying a permanent reconciliation tax — and its dashboards become negotiations rather than instruments.

The trap that looks like speed

A platform that makes work easier to start, and does not cap what is in progress, lengthens lead time rather than shortening it. Little's Law does not stop at the screen: with throughput fixed, more in flight is more time in flight. The platform delivers exactly what it promised — frictionless initiation — and the organisation experiences it as delay.

This is worth saying plainly because it is the most common disappointment in enterprise tooling, and it is almost never diagnosed correctly. The tool is not underperforming. It is performing, and nobody capped the queue.

THREE

Philosophy eats AI

Every digital transformation now passes through artificial intelligence, and AI is the extreme case of everything above: the most expensive asset an organisation can buy, and the one whose return depends most completely on a capability that cannot be bought alongside it.

Schrage and Kiron put the claim sharply. The binding constraint on sustainable value from AI is **not technical but philosophical**. Organisations invest heavily in algorithms, data infrastructure and talent while never articulating what their systems should value, how they should know, and what they should represent about reality. Those assumptions do not stay harmless because they were never written down. They are executed, at scale, continuously.

Three pillars

Each governs a different question, and each has an exact counterpart in what these two sessions have already established — which is the reason to treat them as operations content rather than as philosophy imported for decoration.

Teleology — what the system is told to want

The objectives and the hierarchy among them. It is the most critical of the three and the most frequently skipped, because a single objective is easy to state and a *ranking* of conflicting objectives requires a decision nobody wants to put in writing.

When a system optimises the wrong metric, or when objectives conflict with no explicit hierarchy, it produces outcomes that are technically correct and strategically destructive. A demand forecast optimised to minimise inventory will systematically underestimate seasonal peaks. A credit approval system optimised to reduce default will systematically exclude profitable segments.

> WHERE YOU HAVE MET THIS
>
> Order winners and qualifiers. A system optimising the wrong metric is a system whose purposes were never ranked — exactly the organisation that pours effort into a qualifier above the threshold and wonders why preference did not follow.

Epistemology — what the system may count as knowing

How it acquires knowledge, how it validates it, and how it expresses certainty. Systems that do not separate correlation from causation, that do not quantify uncertainty, or that treat historical patterns as permanent truths are epistemologically fragile.

The practical consequence is decisions taken with high statistical confidence in conditions that no longer resemble the ones that generated the training data. The pandemic invalidated the historical data behind practically every demand model in operation simultaneously — and the models kept reporting confidence throughout.

> **WHERE YOU HAVE MET THIS**
>
> Signalling at the source. A system that knows when it does not know, and escalates rather than answering, is an andon: the abnormality surfaces the moment it occurs instead of arriving in the next monthly report. A system that always returns an answer, with no stated confidence, is aggregated reporting with a faster clock.

Ontology — what the system is able to see

Which entities it recognises, which relations it models, which variables exist for it at all. A sourcing system that represents suppliers by price and quality alone — ignoring geopolitical exposure, sustainability and innovation capacity — has an impoverished ontology. Its recommendations will be locally optimal and systemically fragile.

> **WHERE YOU HAVE MET THIS**
>
> Causal ambiguity, from the benchmarking argument that follows. Representing a supplier by price and lead time alone is the same error as benchmarking the indicator instead of the practice that produces it: in both cases the representation omits the thing that actually explains the result.

Philosophical debt

Schrage and Kiron name the accumulation: **philosophical debt** — unarticulated assumptions about purpose, knowledge and reality, embedded in the models, compounding silently until the damage is hard to reverse. The analogy with technical debt is exact in form, and the difference is where the argument of this day lands.

> Technical debt accumulates in A and P.
> Philosophical debt accumulates where C should be.
>
> a shortcut in the artefact · an absence in the organisation

Technical debt is a compromise somebody made and could document. Philosophical debt *is* the absence of the documentation: nobody wrote down the hierarchy of purposes, so nobody can audit it, so nobody can improve it. An organisation that cannot say what its system is optimising for has no object on which to run an improvement routine. That is a deficit in capability, arriving dressed as a purchase of assets.

THE FAILURE THESE THREE PREDICT

A sourcing system optimised for cost and delivery which, two years on, has systematically excluded suppliers in the regions the firm's own strategy prioritised. The fault is not in the model. The correction is not technical either — adding a column does not resolve it. The hierarchy of objectives was never articulated, so the trade-off was decided by default; and without articulating it, the technical fix simply creates a new debt in the opposite direction.

From predictive to agentic, and why the stakes change

Most AI in operations today is predictive: the system analyses, forecasts or recommends, and a human decides and acts. Agentic systems formulate plans and take action in pursuit of objectives without a human in each step. Along all three pillars, what was documentation becomes a control surface.

Dimension	Predictive	Agentic	What leadership must decide
Autonomy	Recommends; a human decides and acts	Plans, acts, learns from outcomes	Explicitly where autonomy begins and ends — and who is accountable
Teleology	Objectives fixed by whoever configured it	Formulates and refines goals within limits	The hierarchy of purposes becomes the most critical strategic document
Epistemology	Validates against historical patterns	Identifies gaps in its knowledge and seeks new information	Governance of sources and of what counts as valid evidence
Ontology	Fixed representation from the training data	Can extend its representation with new concepts	Periodic review of which variables the system considers — and which it ignores

THE QUESTION FOR THE NEXT STEERING COMMITTEE

Technical safeguards — value limits, audit logs, anomaly alerts — are necessary and not sufficient. Before granting any system autonomy, three philosophical ones:

Teleology: is the hierarchy of purposes documented, including what the system should do when cost, quality and sustainability conflict? **Epistemology:** does it know when it does not know, and escalate below a confidence threshold? **Ontology:** can it recognise a supplier or a context it has never seen, or does it only optimise inside the world it was trained on? Without those three, autonomy is risk wearing the costume of efficiency.

FOUR

From a working model to an organisational result

Part Three argued that where a tool contains AI, the binding constraint is conceptual rather than technical. That is an argument. This part is the measurement — and it comes from the only studies that installed AI in a real organisation, changed something, and measured what happened afterwards.

PROVENANCE, AND THE LIMIT OF THE EVIDENCE

The framework in this part is drawn from a systematic review by the author and Claudia Araujo (COPPEAD/UFRJ, manuscript in preparation), conducted under PRISMA 2020 across four databases. **Its evidence base is hospital implementations.** The six conditions are stated here in general organisational terms, which is an extension of the framework and not a finding of the review; every figure quoted below is a hospital figure and is labelled as such. It is offered as a structure for appraisal, not as a citable result.

Most of the literature stops at the model

Of 1,271 records screened, **37** had both an implementation and a comparison — roughly one for every eight that end at a model or a simulation. The largest single exclusion category, 464 studies, reported algorithmic metrics only: accuracy, AUROC, F1. They are studies about a model, not about an organisation.

That ratio is the first finding, and it travels well beyond healthcare. The published evidence that an AI tool works is overwhelmingly evidence that it predicts well. Whether anything changed in the organisation is a different question, far less often asked, and the answer is not implied by the first.

WHAT THE LITERATURE MEASURES, AND WHAT THAT IMPLIES

Of the 37 studies, **36 measured productive efficiency** — time, cost, throughput. Patient-centredness appeared in 8, responsive governance in 2. Read through the morning's arithmetic, this is a striking result: AI as actually implemented and measured is almost entirely an exercise in lowering cost, which is the edge of value where every competitor is also working. It is a *qualifier* programme being argued as a winner. VC, the difference between two firms, is not where this literature looks.

Six conditions, each necessary, none sufficient

Organisational value is the end of a chain. A break at any link stops the effect, however good the preceding links were. The two middle conditions are not properties of the tool at all: they are capabilities of the organisation that bought it.



Conditions of the tool, and of the result

The two organisational capabilities — this is C, made measurable

	Condition	The question to ask	The failure signal observed in the review
0	**Evidence base**	Has this class of application produced a measured result in a real organisation — not in a model and not in a simulation?	Many modelling studies, no measured implementation; divergent results inside the same application category.
1	**Technical capability in operation**	Does the model still perform on our data, at the threshold actually in use, counting only outputs that arrive before people had already noticed?	Performance reported before deployment not reproduced in routine use — and the organisation unchanged even where the model stayed accurate.
2	**Workflow integration**	Does the output reach whoever decides, while the decision is still open? Who receives it, at which step, and is it in fact used?	Output ignored by most of the staff, or arriving after the decision had been taken.
3	**Translation into action**	When the output arrives, is there an owner able to interpret it and act — with the resources and dependencies that action requires?	Cases correctly identified and no capacity to act on them: people never contacted, actions blocked on unavailable resources.

4	**Organisational outcome**	Which management indicator should move, from which data source, against which baseline — measured without relying on opinion?	Effect concentrated in a single performance dimension; perceived gains not confirmed by objective measurement.
5	**Net value**	What will the gain cost, and who bears that cost? Who is de-prioritised, what work is displaced, what does verification take?	Gains offset by work shifted to other people or other hours, and by the effort of checking generated content.

Used **before** adoption, the chain is a checklist of conditions the organisation itself controls. Used **after**, it locates the link where the implementation failed — which is a different and more useful question than whether the model was accurate.

> CONDITION 4 HAS A NAME IN THE PROJECT LITERATURE
>
> Naming the indicator, its data source and its baseline before approval is **benefits realisation management**, and Serra and Kunc (2015) find that the practices which govern it are associated with project success and with the execution of business strategy — not with better delivery of outputs, which is the distinction that matters here. A digital investment appraised without it can be delivered on time, on budget, to specification, and still leave nothing to point at afterwards. Condition 4 is that finding written as a question somebody has to answer in writing.

Model accuracy does not separate success from failure

In the studies with no effect or an unfavourable one, predictive performance stayed high and stable wherever it was measured. Four mechanisms in the fit between tool and work explain them, and they group cleanly into the two middle conditions. The figures are from hospital implementations.

Mechanism	Condition	What was observed
Adoption	2 · integration	**67%** of staff rarely or never used an admission-prediction tool. Adjusted odds ratio for the implementation period: 1.03 (95% CI 0.93–1.13). The model was fine. Nobody opened it.
Timing	2 · integration	A neural network's output reached the physician *after* the cardiac markers returned, when the disposition decision had already been made. It changed management in **2 of 432** cases.
Reach	3 · translation	**42.3%** of the patients the model flagged as likely no-shows were never contacted. In a separate nodule follow-up, 47 of 80 were not located, with delays attributed to staffing.

Execution barrier	3 · translation	Time fell for low-barrier actions and not for actions requiring a chain of dependencies. Earlier warning does not accelerate an action whose constraint was never information.

THE COUNTERPOINT, WHICH IS THE USEFUL HALF

Where the intervention operated on work that was **already centralised**, the effect appeared. The same model engaged pharmacists, whose work was centralised, while the component requiring a new, decentralised behaviour failed in the same hospital. Elsewhere the useful element was not the recommendation engine but the prioritisation, because it triggered a discussion inside a routine that already existed. The practical rule: an AI output lands where there is already a routine to receive it, and dies where it would have to create one.

Two studies that settle the argument

Two randomised studies of appointment no-shows share the application, the action — a telephone call — and the design, and reached opposite results.

Study	Result	Where the chain stands
McMahon et al., 2026 MRI no-shows, Switzerland	21.9% vs 22.0% (p = 0.48). No effect.	Model discrimination had drifted to a modest AUROC of 0.61, but the output was tied to a concrete action. The chain broke at **reach**: 42.3% of those selected were never called.
Dunstan et al., 2023 paediatric no-shows, Chile	21.0% → 10.7%, −10.3 points against a concurrent control (p ≈ 0.002).	Worked — and the contact rate was never reported. The chain identifies the **missing measure**: we cannot say why this succeeded where the other failed.

A model-centred appraisal cannot tell these two apart, because on model terms they are the same study. The difference lies entirely in condition 3, and it was measured in one case and not in the other.

Three conditions on the evidence itself

These are not links in the chain. They qualify whatever evidence reaches the decision-maker at every link, and each has a counterpart in how vendors present results.

- **A concurrent comparator.** Before-and-after with no control attributes to the tool everything else that changed in the same period. Nineteen of the 37 studies had no concurrent group.

- **Objective measurement.** Where perception and objective measurement were collected in the same study, they diverged: in one, search took 43 seconds *longer* with the tool, and users reported that it saved them time.
- **Independence from the vendor.** A vendor-reported sensitivity of 91% fell to 18% once restricted to alerts arriving before clinical recognition — the only alerts that could have changed anything. Notably, the 13 studies declaring vendor ties were *less* favourable than the rest, which is the opposite of sponsorship bias and suggests the optimism is upstream of the funding.

The laboratory-to-floor gap is the general form of all three. A contouring tool that saved 93% of the time in the laboratory saved 36% to 67% in the clinic. Nothing about the model changed; the setting did.

HOW THIS JOINS PART THREE

The three pillars and the six conditions are the same argument at different altitudes, and each pillar fails at an identifiable link. **Teleology** fails at conditions 4 and 5 — a system with no documented hierarchy of purposes produces a gain on one indicator and a cost on another that nobody assigned. **Epistemology** fails at condition 2 — a system that does not know when it does not know sends an output nobody can act on, at a moment when nobody can act. **Ontology** fails at conditions 1 and 3, where the world the model was trained on meets a case it has no category for.

Philosophical debt, read this way, is not an abstraction. It is a measurable break in a chain, and conditions 2 and 3 are where it surfaces first.

IN PROJECT TERMS

Conditions 2 and 3 are almost never in scope. A digital project is scoped to deliver a working system — condition 1 — and the integration into someone's decision and the capacity to act on the output are treated as adoption, change management, or somebody else's problem after go-live. The review says that is where the value is won or lost.

The practical consequence for a business case: name the decision the output enters, the person who owns the resulting action, and the resources that action consumes — before approval, not after. A project that cannot name all three has not identified a benefit; it has identified a capability, which is condition 1 and is not the same thing.

Adopting AI is a process-design decision, not a technology purchase.

conditions 2 and 3 are the organisation's, not the vendor's

FIVE

Benchmarking: Decoding the rules-in-use

Benchmarking is conventionally practiced as corporate espionage or tourism: visiting high-performing firms, photographing their tools, buying their software, and pasting their org charts. That is why it fails.

The method, before the critique

Benchmarking has an established methodology, and the criticism that follows in this part is a criticism of how it is practised rather than of the method itself. Camp's formulation, from the Xerox programme that made the practice general, runs in four movements.

	Phase	What it requires	Where it is usually skipped
1	**Plan**	Decide what to benchmark — a process, not a department — identify a comparator, and settle how the measure will be defined before anybody goes looking.	The comparator is chosen for prestige rather than for comparability.
2	**Collect**	Gather data on both sides using the same definition, and record the conditions under which each figure was produced.	Their figure is taken from a published source; ours is calculated differently.
3	**Analyse**	Establish the size of the gap and, far more importantly, the practice that produces it.	The gap is quantified and the practice is never identified. This is the fatal one.
4	**Adapt**	Set targets, change the practice, embed it, and re-measure.	Targets are set from the gap with no change to the practice that would close it.

Choosing the comparator

Four kinds, in increasing order of difficulty and of payoff.

- **Internal** — another unit, plant or region in the same organisation. Easiest access, most comparable conditions, smallest ceiling: it can only take you to the best you already are.
- **Competitive** — a direct rival. The most wanted and the least available, and the one most likely to converge the industry, for the reasons set out below.
- **Functional** — an organisation that performs the same function in a different industry.

- **Generic** — a process that is structurally analogous in a wholly unrelated setting. Xerox famously fixed its warehouse by studying a catalogue retailer rather than a competitor, and the distance is what made the practice visible instead of invisible.

Making the numbers comparable, and using international comparators

Most benchmarking failures are arithmetic before they are strategic. Three corrections do most of the work: **define the boundaries** of the process identically on both sides — most on-time-delivery disputes are disputes about when the clock starts; **normalise for scale and mix**, since cost per unit across different volumes and product mixes is not one measure but two; and **record the conditions**, because a figure achieved under different regulation, labour markets or demand variability is not a target, it is a datum.

International comparators — standards bodies, professional-association survey data, published maturity studies — are useful for exactly one thing: establishing whether a gap exists at all, and roughly how large. They cannot tell you what produces it, because they report distributions rather than practices. Treating a percentile from a survey as an improvement objective is the error in phase 3 above, formalised and given a chart.

Why the copy-paste approach guarantees mediocrity

Returning to the NUMMI puzzle from Session 1: General Motors took tours of NUMMI for twenty years. They photographed the kanban boards, copied the andon lights, and replicated the physical workstations in their other plants. The other plants remained unproductive and combative.

Why? Because GM copied **Assets (A)** and documented **Processes (P)**. They could not observe, let alone import, the tacit **Capabilities (C)** and the *Rules-in-Use*.

$$V_{rival} = C_{weak} \times (A_{bought} + P_{copied}) \approx 0$$

The mathematical explanation of failed benchmarking

Why best practice resists transfer: the evidence

NUMMI is not an anecdote about one stubborn company. It is the most visible instance of a finding that strategy and operations research reached independently, and the literature explains the mechanism with unusual precision. Four results matter.

It fails even inside the same firm

Szulanski (1996) studied the transfer of best practice between units of the *same* organisation — no competitive barrier, shared incentives, common systems, cooperative intent. Transfer stalled anyway. The dominant obstacles were not motivational, which is what managers assume; they were knowledge obstacles: **causal ambiguity** about why the practice works, limited **absorptive capacity** in the receiving unit (Cohen &

Levinthal, 1990), and a difficult relationship between source and recipient. He named the phenomenon *internal stickiness*.

The implication is severe. If a practice will not travel reliably between two plants of the same company, assuming it travels from a competitor through a site visit and a slide deck is not conservative. It is heroic.

Observation is not understanding

Polanyi (1966) set the base case: we know more than we can tell. Nelson and Winter (1982) located organisational knowledge in **routines** — repeated patterns of coordinated action that are largely tacit and that no document fully specifies. Lippman and Rumelt (1982) named the strategic consequence: under **causal ambiguity**, even the firm that *owns* a superior practice may not know precisely which elements produce the result. It therefore cannot reliably teach it, and a rival cannot reliably copy it.

GM's visitors were not looking at the wrong things. They were looking at the only things that could be looked at.

Partial copies perform worse than no copy

Rivkin (2000) showed formally that when a strategy is a *system of interdependent choices*, imitation stays hard even if every element is visible. The number of possible combinations explodes, and a rival who gets most elements right and a few wrong typically performs **worse than one who copies nothing at all**, because the mismatched elements work against each other.

This is the mathematical form of the complementarity argument from the morning, and the precise reason GM's other plants — kanban boards installed, andon lights fitted, the rest absent — got worse rather than better.

Capability is accumulated, not purchased

Dierickx and Cool (1989) separated assets that can be bought on a market from asset *stocks* that must be accumulated over time, and identified **time compression diseconomies**: building in one year what a rival built in five costs disproportionately more, and often cannot be done at any price. Winter (2003) then drew the distinction that organises this whole session: an **operational capability** is how the firm earns its living today; a **dynamic capability** is its capacity to change the operational ones.

In the language of the morning: *P is the operational capability. C is the dynamic one.*

Mechanism	What it predicts	What it changes in your study
Internal stickiness Szulanski (1996)	Transfer fails for knowledge reasons, not motivational ones	Budget for absorption, not only for acquisition. Who here can receive this, and what must they already know?
Causal ambiguity Lippman & Rumelt (1982)	Even the source may not know which elements matter	Do not ask the reference what it does. Ask what it stopped doing, and what it would never give up.

Interdependence Rivkin (2000)	Partial imitation underperforms no imitation	Identify the bundle, not the practice. If you cannot adopt the set, adopting one element may cost you.
Time compression Dierickx & Cool (1989)	Accumulated capability cannot be bought quickly at any price	Ask how long the reference took. Then plan for that order of magnitude, or change the objective.

THE RULE THIS GIVES YOU

Do not benchmark the number. Benchmark the routine that produces it — then ask what your organisation would have to accumulate in order to run that routine. The first question has an answer you can put on a slide. The second has an answer you can act on.

THE STRATEGIC CAVEAT

There is a second-order risk the literature above does not address, and Porter (1996) does. Benchmarking is by construction an *imitative* activity: it moves the organisation towards a frontier others already occupy. Practised well across an entire industry, it converges competitors onto the same operating model, and competition collapses into price.

The morning gave this its arithmetic. Competitive advantage was written as a difference — $VC = (WTP_A - Cost_A) - (WTP_B - Cost_B)$ — and a successful imitation is exactly the event that pulls the second bracket up towards the first. Benchmarking is therefore the one improvement activity whose logical endpoint, when both sides practise it well, is VC equal to zero for everyone in the industry.

That is not an argument against benchmarking. It is an argument for being explicit about which of two things you are doing — closing a gap on a qualifier, where convergence is precisely the goal, or building an order winner, where imitation is the wrong instrument entirely.

Spear & Bowen's Four Rules applied to Digital Benchmarking

The theory says what to look for. Spear and Bowen (1999) say where to look. Their four rules are the decoding key — the tacit operating rules a site visit will never surface unless the visitor arrives knowing they exist.

Rule	What naive benchmarking sees	What rigorous benchmarking decodes

1. Standardized Work	"They have standard operating procedures and software forms."	Work is specified to the level of content, sequence, timing, and direct observable outcome. Any variance is treated as a testable scientific problem.
2. Direct Connections	"They use a central ticketing platform or Slack channel."	Every customer-supplier connection within the organization is binary, direct, and unambiguous, with zero intermediate translation or ambiguous handoffs.
3. Simple Pathways	"They have automated end-to-end routing in their ERP."	Every product, service, or safety signal travels along a pre-specified, single pathway. No work loops back or diverts unless explicitly designed.
4. Scientific Method	"They run monthly retrospective meetings."	Any improvement must be structured as a refutable hypothesis, mentored by a senior leader, and executed at the lowest possible organizational level.

SIX

The transformation vector — how this is actually run

Everything so far has been diagnosis: what the money bought, where the chain breaks, why the copy underperforms the original. Diagnosis ends on a question no business case answers. If the path cannot be known in advance, what replaces the plan?

This part is drawn from Chapter III of *Excelência Operacional*, where Cynefin, Lean, Lean Startup and Toyota Kata are read as one system. The order matters: Cynefin says which method the problem admits, and Kata is the routine that executes it.

The domain decides the method

Cynefin — Welsh for *habitat* — is a sense-making instrument. Its premise is that different contexts require fundamentally different styles of decision, and that the characteristic error of leadership is not a bad decision but a decision made with the method of the wrong domain.

Domain	Nature of the problem	Approach	The trap
Simple	Predictable and stable; cause and effect obvious to everyone	Sense → Categorise → Respond. Apply the established protocol	Rigidity and complacency; blindness to a change of context
Complicated	Requires expertise and analysis; several right answers exist	Sense → Analyse → Respond. Consult specialists, analyse data	Paralysis by analysis; excessive dependence on specialists
Complex	Cause and effect visible only in hindsight; patterns emerge from experiments. **Innovation, culture and digital transformation live here**	Probe → Sense → Respond. Safe-to-fail experiments, iterative learning	Wanting full control before acting; planning what can only be discovered
Chaotic	No perceptible causal relation; speed matters more than precision	Act → Sense → Respond. Immediate action to stabilise	Inaction, or analysis during a crisis
Confusion	The leader does not know which domain they are in. The most dangerous state	Break the problem up and reclassify each part	Applying a preferred decision style with no diagnosis at all

Most leaders are trained for the Complicated domain, and that training serves a large share of daily operational problems well. The difficulty is that the highest-stakes problems — disruptive innovation, digital transformation, cultural change — are inherently Complex. Applying Complicated methods there is not merely ineffective: it is counterproductive, because a detailed plan manufactures a sense of control that delays experimentation and penalises learning.

Netflix and Blockbuster, 2007

In 2007 there was no best practice available to establish whether consumers would pay for streaming. Netflix operated the problem as Complex: it launched streaming as a free benefit for DVD subscribers (**probe**), observed adoption patterns (**sense**), and only then pivoted the business model (**respond**). Blockbuster treated the same question as Simple and applied historical best practice.

The same market, the same moment, two opposite readings of the domain. One of the two firms was gone by 2010.

WHY THIS INDICTS THE PREVIOUS PART

Benchmarking is a Simple-and-Complicated instrument. It presupposes that a knowable right answer already exists somewhere and that the task is to locate and transfer it. Pointed at a Complex problem it does something worse than fail: it imports a confident answer to somebody else's question, and the confidence is what does the damage. That is the mechanism behind Part Five's finding that a partial imitation can underperform not imitating at all.

Kata: the routine that makes probe–sense–respond repeatable

Probe → Sense → Respond is a logic, not a practice. Without a routine it is indistinguishable from improvisation, and it decays the moment the urgency passes. Kata — in the martial arts, a pattern rehearsed until it becomes reflex — is a routine of scientific thinking applied daily. It does not prescribe what to do. **It prescribes how to think.**

	Step	What it requires	Horizon
1	**Direction or challenge**	Aspirational enough to orient, concrete enough to guide daily decisions. The true north that replaces micromanagement with shared purpose.	1–3 years

2	**Grasp the current condition**	Go to the gemba and observe the process as it actually is. Patterns, gaps and facts — and the knowledge is shared, so the next cycle does not restart from zero.	Continuous
3	**Set the target condition**	A measurable future state, neither trivial nor impossible: in the zone of productive learning, which demands effort without paralysing.	Up to 3 months
4	**Run experiments**	PDCA daily. The path is deliberately obscure — were it clear, there would be nothing to learn. Each experiment moves the boundary of what the organisation knows.	Weekly / daily

> **THE COACHING ROUTINE — FIVE QUESTIONS**
>
> The mentor does not supply the answer; the questions develop the capacity to find it. **1.** What is the target condition? **2.** What is the current condition? **3.** What obstacles are in the way, and which one are you addressing now? **4.** What is your next step, and what do you expect to happen? **5.** When can we see what you learned from that step?
>
> The point of the five questions is not to solve the problem. It is to produce people who solve problems — which is the only form in which capability survives the person who had it.

This is also the answer to the most common failure of Lean programmes. Tools are adopted — kanban, 5S, value stream maps — and the improvement does not survive six months. Kata's explanation is that the tools were the visible content and the routine was the form that sustained them: no mentor asking the five questions, no active PDCA cycles, no defined target condition. The tools get used while the urgency lasts and are abandoned when it passes.

Why the routine builds the coefficient

Kata's claim on C is specific, and it is the claim that connects this part to the close of the day. Organisations have excellent principles and poor routines, and **it is the routines that determine the real culture — not the values on the wall.**

Level of learning	How the routine produces it	What accumulates
Single-loop adaptive	Daily PDCA in the improvement kata: each experiment adjusts the action without questioning the governing norms	Operational efficiency, compounding over time
Double-loop generative	The coaching routine creates the safety to question the status quo: the learner is expected to investigate the facts without pressure	Revision of the norms themselves, and of leadership's mental models

Deutero-learning learning to learn	The coaching routine in its essence: it teaches a method for any problem, not the answer to one problem	Self-renewal — the most durable form of the coefficient

The three levels are Argyris and Schön's; the mapping onto Kata follows Michels and Forcellini (2023), who also note that empirical evidence for the link across varied contexts remains scarce. The second level is worth dwelling on, because it is the one that explains a cultural prerequisite rather than asserting it: double-loop learning requires questioning the governing norms, which an organisation that punishes error cannot do. People conceal mistakes and avoid risky experiments, and they are behaving rationally. *Yokoten* — circulating know-how across the organisation without restriction — is the complementary mechanism: it is what turns an individual's learning into collective property, and therefore what arrests the outflow Part Eight describes.

> **WHERE THIS MEETS ANOTHER SESSION OF THE MODULE**
>
> The experiment logic below overlaps deliberately with the Design Thinking session earlier in Module 3, which covers prototyping and rapid experimentation in their own right. It is repeated here for one reason: to connect the experiment to a *governance* question — what an early-stage initiative can honestly take to a finance committee — and then to hand it to the Kata routine, which is what keeps experimentation running after the workshop energy has gone.

When the path is unknown, what do you take to the CFO?

The Complex domain creates a governance problem, not only a method problem: an early-stage initiative cannot produce the ROI projection that approval requires. The financial argument is sound and rarely made. Early-stage innovation ROI is by definition zero or negative — exactly like any asset under construction that does not yet generate cash flow. The real exposure is not the provisional negative return; it is the risk of building something for which there is no demand, and that risk is only reduced by early validation. Validated learning is therefore best presented as **capital-risk reduction**: each hypothesis confirmed raises the probability of a positive return on the total investment.

	Phase	How to execute it	The trap
1	**Frame the hypothesis**	Identify the one assumption which, if false, invalidates the whole model. State it in a single testable sentence.	Vague hypotheses, or several at once, which make diagnosis impossible
2	**Build the MVP**	The leanest version that tests that assumption. In services this can be a spreadsheet, a flyer, or fifteen minutes of conversation. The object is to learn, not to impress.	An over-elaborate MVP — confusing a product with an experiment
3	**Measure the learning**	Decide *before* the experiment which signal confirms or refutes the hypothesis. Behavioural evidence, not stated intention.	Vanity metrics: activity in place of real buying or using behaviour

4	**Pivot or persevere**	Confirmed: scale. Refuted: change one fundamental variable and start again. Speed here is itself an advantage.	Persevering from inertia, pivoting from impatience — both without evidence

THE DILEMMA THIS PART EXISTS TO ANSWER

A competitor implements generative AI in its processes and cuts cost by 30%. What do you do?

1 · Cynefin. Is the thing in front of you Simple — copy the technology — or Complex — discover how AI could create value specific to your customer? Do not confuse the two. **2 · Lean.** Is human contact a driver of willingness to pay in your segment? If it is, automating it cuts cost and destroys revenue at the same time. **3 · Lean Startup.** Before deciding, build an MVP: test on a subset of customers and measure the effect on NPS and churn, not only on cost. **4 · Kata.** Set a 90-day target condition and run controlled experiments towards it.

The reflex to copy the competitor usually reproduces operational effectiveness without producing differentiated advantage — which, in the morning's notation, is an improvement that leaves VC where it was.

IN PROJECT TERMS

This is the stage-gate problem stated precisely. A gate asks for a plan; a Complex problem can only supply a hypothesis, so the plan gets written anyway and the organisation then manages the plan instead of the problem. The reconciliation is not to abandon governance but to change what the gate receives: the deliverable of an early gate is a target condition and the evidence from the last experiment, not a schedule and a projection.

Ries's portfolio framing is the governing discipline — mature operations and high-risk innovation managed as different asset classes, with different metrics and different tolerances. Mixing the two logics kills promising projects early and sustains stagnant ones indefinitely.

SEVEN

Structured activity — the Digital Tool Appraisal

The morning asked whether a project was worth funding. This afternoon asks a harder question about the same project: of the digital tooling it depends on, what is actually being bought — and what is being left undone?

> **THE INSTRUMENT**
>
> The appraisal is a separate page, fillable and exportable in the same way as the morning's canvas: **Digital Tool Appraisal**. Open it now. It carries its own instructions; what follows is why it asks what it asks.

Four layers, and why they are one instrument

The appraisal does not inventory features. A list of whether the tool has a poka-yoke, how many systems people toggle between and whether an andon exists is an audit, and an audit is work for one person with a checklist — not for four senior managers with twenty minutes. What four senior managers can do that a checklist cannot is **judge**, and each layer forces a judgement that is uncomfortable to make honestly.

Layer	The judgement it forces	Where it comes from
VCAP triage	Does this change what we own, how work flows, or what we are able to improve? Most honest answers land on the first two.	This morning's identity, V = C × (A + P)
Philosophical debt	Where the tool contains AI: has anyone written down what it should want, what it may count as knowing, and what it is able to see?	Part Three — Schrage & Kiron's three pillars
The six conditions	Where the tool contains AI: at which link does the chain from a working model to a result currently break, or go unmeasured?	Part Four — the six-condition chain
The reference	What does somebody already do that we do not — the routine, not the number — and what would have to be true here for it to work?	Part Five — the method, then internal stickiness, causal ambiguity and time compression
Maturity and roadmap	Which capability gaps most constrain performance, and in what order can they credibly be closed over six months?	The six conditions read across, plus strategic value against feasibility

The fifth layer is where the instrument stops being about one tool. Read across, the six conditions are a maturity statement about the organisation — which conditions it can satisfy and demonstrate, which it satisfies without evidence, and which it has never observed. The instrument deliberately produces no composite maturity score: a single figure would conceal the only thing worth knowing, which is *where* the chain stops. The roadmap then sequences the three largest gaps by strategic value against feasibility, which is the output the module's assignment asks for.

The two AI layers do different work and the instrument keeps them apart deliberately. The pillars ask whether anything was *articulated*; the six conditions ask whether anything was *measured*. A tool can pass the first and fail the second — a clearly stated purpose whose output reaches nobody in time — and it can pass the second and fail the first, which is the more dangerous case: a chain that demonstrably moves an indicator, with no one able to say which other indicator is paying for it.

> **WHERE THE VERDICT LEADS**
>
> A verdict is not a score, and the instrument does not produce one. It produces a located gap — and Part Six says what to do with it. If the gap is at condition 2 or 3, the problem is Complex: the integration into somebody's decision and the capacity to act cannot be specified in advance, so the next move is a target condition at ninety days and an experiment, not a requirements document. If the gap is a missing measure, the next move is simply to measure it before spending anything further.

What to expect

"Not documented" will be the honest answer

Most groups find that nobody can say what the system optimises for when objectives conflict. That is the finding, not a failure of the exercise — and it is cheap to fix before deployment and expensive afterwards, once the organisation has two years of decisions made under the implicit hierarchy.

Most tools will land on A, or A and P

That is a correctly identified purchase, not a weakness. What it means is that a competitor with the same budget can hold the same position — so the business case should be argued on that basis rather than on advantage.

Expect, too, that the six conditions will be answered *not measured* more often than *no*. That is the characteristic result and it is worth saying out loud: an organisation that cannot say whether the output reached the decision in time, or whether the recommended action was ever carried out, does not have a failing implementation — it has an unobserved one, which is a different and more tractable problem.

Bring the exported appraisal with you. Together with the morning's Value Case it is the input to the module's Execution Optimization Challenge.

EIGHT

Building the multiplier

The day has been an argument by elimination. Assets can be bought. Processes can be documented and taught. Benchmarking transfers the second and purchases the first, and neither moves the term that multiplies them. One question is left, and it is the only one with a practical answer: **how is C actually built?**

Capability is a stock, not a state

Dierickx and Cool's distinction, from Part Five, organises everything here: flows adjust instantly, stocks do not. Treating the capability to improve processes as a stock means asking three separate questions — what flows in, what flows out, and what conditions regulate both.

	Mechanism	Managerial consequence
Inflow	Deliberate improvement activity that produces causal understanding and validated action together	Volume of improvement projects is not the measure. Projects that produce only one of the two do not accumulate anything.
Condition	Psychological safety and the credibility of management's commitments	Where surfacing a problem is costly, the measured performance improves and the real performance does not.
Outflow	Turnover and disuse	An organisation that was good at improving five years ago may no longer be, and nothing on its balance sheet will say so.

What the inflow actually requires

The morning established the finding: across improvement projects in a single factory, only about a quarter accelerated the reduction of waste — the ones producing *know-why* and *know-how* simultaneously (Lapré et al., 2000). The rest slowed it or had no effect. Three mechanisms follow from that, and the operations literature describes them with increasing precision.

- **Guided experimentation at the point of execution.** Improvement run as a hypothesis, tested by the people doing the work, under a more experienced coach, at the lowest level where the problem is visible. This is Spear and Bowen's fourth rule from Part Five, now read as an accumulation mechanism rather than as a benchmarking target — and it is the best-documented inflow in the literature. **Part Six gave it a name and a cadence:** the improvement kata is this mechanism operationalised, and the coaching kata is what keeps it running when nobody is being audited.

- **Team learning.** Capability lives in coordinated routines, not in individuals. A team that cannot discuss a failure cannot refine a routine, which is why psychological safety is a precondition rather than a nicety. In the terms of Part Six, it is the precondition of double-loop learning: an organisation that punishes error can run single-loop PDCA indefinitely and never revise the norm that produced the problem.
- **Credibility.** Sustained improvement effort depends on management's promises being believed, above all the promise not to convert improvement gains into redundancies. People do not improve away their own jobs, and they are right not to. This is a condition, not a value statement: break it once and the inflow stops.

> **THE COST OF TRYING TO GO FASTER**
>
> Time compression diseconomies apply to this stock as they do to any other. Doubling the improvement effort does not halve the time to accumulate, and past some point the additional effort returns less than it costs. An organisation that wants in eighteen months what a reference built over a decade is not facing a budget problem. It is facing an arithmetic one, and the honest response is to change the objective rather than the budget.

> **IN PROJECT TERMS**
>
> Every element above has a project-management equivalent that most organisations already own and few use as intended. The retrospective is guided experimentation — if its output changes the standard, and nothing if it does not. The lessons-learned repository is an attempt to arrest the outflow, and it fails for exactly the reasons Part Five gives: it stores what can be written down, which is the part that was never the constraint.
>
> The practical test for a PMO: after the last three projects closed, what can this organisation now do that it could not do before — that does not depend on the people who happened to be on those teams?

Design is not the same as practice

Everything above describes something an organisation can *design*: which capability it sets out to build, which assets it buys, which processes it writes down, which improvement routine it adopts. All of that is specifiable, auditable, and — the whole lesson of the afternoon — visible to a competitor. Design is necessary and it has never been sufficient, because the same design produces different results in different organisations. That gap has a name, and a form:

$$\text{Operational Excellence} = \text{Design(VCAP)} \times \text{Culture}$$

what the organisation specified · multiplied by whether it is actually lived

The formulation above is the author's adaptation. The structure it borrows — a designed solution *multiplied* by its acceptance in practice rather than added to it — is long-standing in the literature on organisational change; the reading of the first term as a VCAP configuration is this course's.

Design(VCAP) is the deliberate configuration: the chosen capability, the assets acquired, the processes written. It is the blueprint, and it is the part that can be presented, bought, certified and copied.

Culture is the substrate the whole configuration sits on — not a fourth term beside A, P and C, but the condition under which any of them delivers what it was designed to deliver. It is what the organisation does when the auditor has left: whether the standard is followed when it is inconvenient, whether the andon is pulled when pulling it is embarrassing, whether the improvement routine survives a quarter in which the numbers are bad.

> **WHERE C ENDS AND CULTURE BEGINS**
>
> **C is inside the design; Culture is whether the design is real.** C is a capability the organisation chose and can describe — "we are able to improve this class of process repeatedly". Culture is not described, it is observed, and it governs every term at once: an asset used as instructed and an asset worked around, a documented process followed and the same process bypassed, a declared improvement capability exercised and one that exists only in the handbook. That is why it multiplies the expression rather than sitting inside it.

The form is multiplicative for the same reason VCAP is. An addition would say that a strong design partially compensates for an organisation that does not live it — that excellent processes plus indifferent practice yield moderate excellence. They do not. They yield binders. At the limit the statement is blunt: where Culture is zero, Operational Excellence is zero, whatever was designed and whatever was spent.

> **WHAT THIS FORMULATION IS, AND IS NOT**
>
> It is a didactic statement of a relation, not an instrument. Culture has no unit here and no score, and any attempt to supply one would be inventing precision the concept does not carry. The purpose of writing it as a product is to forbid an addition — to make it impossible to argue that a good design and a weak practice average out to something acceptable. Used as a reminder it is sound; used as a measurement it is not.

Where the day ends

$$V = C \times (A + P)$$

A is bought · P is copied · C is the only term you have to build

This morning opened on a plant in Fremont where the assets and the people stayed the same and the result inverted. The afternoon has given that observation its mechanism: what changed was a stock that no visitor

could photograph, no contract could transfer, and no platform could install — and which, left alone, would have drained away again within a few years. Read through the second formulation, Fremont is the cleanest experiment available: the design changed, and so did the substrate it ran on, and only the second explains why the same plant and the same workforce produced a different outcome.

> **WHERE TO READ MORE**
>
> The accumulation mechanisms, the measurement problem, and the conditions under which the identity V = C × (A + P) would be refuted are developed at length in *O que não se copia: capacidade, práticas e desempenho em operações* (Lima, manuscript in preparation), which treats the argument of these two sessions as a set of testable propositions rather than as a framework.

NINE

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Sorbonne Business School · IDGP Academy · Prof. Dr. Thiago Lima